Project - Low Level Design

On

IaC Provisioning for Telecomm System

Course Name: Devops

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2022-2026

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1. Introduction

This document presents the system design for the IaC Provisioning for Telecomm System project, developed as part of the Datagami Skill Based Course at Medicaps University. The project demonstrates the automation of infrastructure provisioning for a Telecom-grade environment using Infrastructure as Code (IaC) tools including Terraform and Ansible.

The objective is to replace manual infrastructure setup with automated, repeatable, error-free provisioning that can deploy both local Docker-based telecom service environments and cloud infrastructure (AWS). The system provisions compute instances, installs dependencies, configures telecom network components, and deploys required services using automated scripts.

1.1 Scope of the Document

This document covers:

- Complete IaC system architecture and provisioning workflow
- Terraform configuration for cloud resource provisioning
- Ansible playbooks for configuration management
- Docker-based local telecom environment deployment
- CI/CD automation for provisioning
- Security considerations and environment variable handling
- API catalogue (if applicable for telecom services)

Excluded:

- Internal telecom business logic
- Proprietary telecom signalling algorithms
- Network protocol handling of telecom core elements

1.2 Intended Audience

Audience	Purpose
DevOps Engineers	Understand IaC automation using Terraform and Ansible
Cloud Architects	Review provisioning, security, and scaling considerations
Telecom Engineers	Understand automated telecom environment deployment
Academic Evaluators	Technical depth and implementation analysis
Future Maintainers	Extend provisioning scripts safely

A basic understanding of Terraform, Ansible, AWS, and Docker is recommended.

1.3 System Overview

The system consists of two decoupled repositories:

Repository 1 — IaC Scripts (Telecomm-IaC-Provisioning)

Contains Terraform modules, Ansible playbooks, inventory files, Docker environment setup, and automation scripts.

Repository 2 — Telecom Application Repository

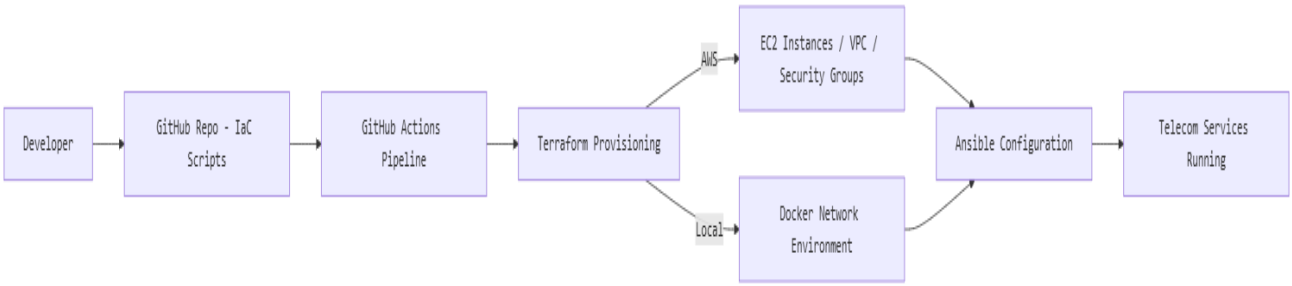
Contains telecom simulation services (e.g., Call Simulation, SMS Router, Signalling Handler, Network Registry).

Workflow Overview

Whenever infrastructure provisioning is triggered:

1. Terraform provisions cloud infrastructure or local Docker networks.
2. Ansible configures OS, installs dependencies, deploys telecom services.
3. Docker containers simulate telecom components locally or on EC2.
4. CI/CD automates the entire provisioning pipeline.

High-Level Architecture Diagram



2. System Design

2.1 Application Design

The system provisions two environments:

- Cloud Environment (AWS) via Terraform
- Local Docker Telecom Network via Docker Compose

Telecom Components in Application Repo

Module	**Purpose**
Call Handler	Simulates call signalling
SMS Router	Routes messages
Network Registry	Stores subscriber metadata
Monitoring Dashboard	Streamlit/Grafana-based UI

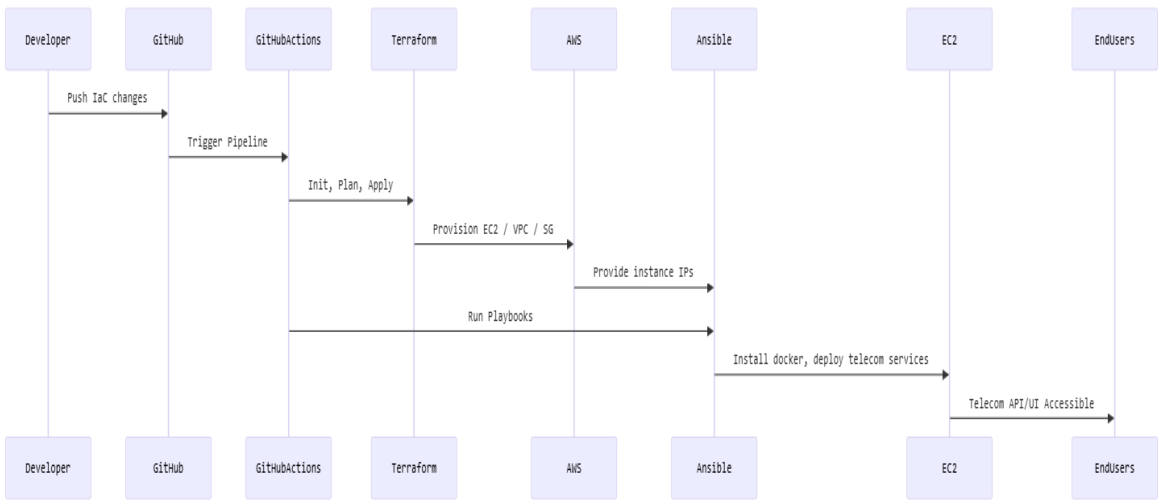
Container Design

Each telecom component packaged into Docker images:

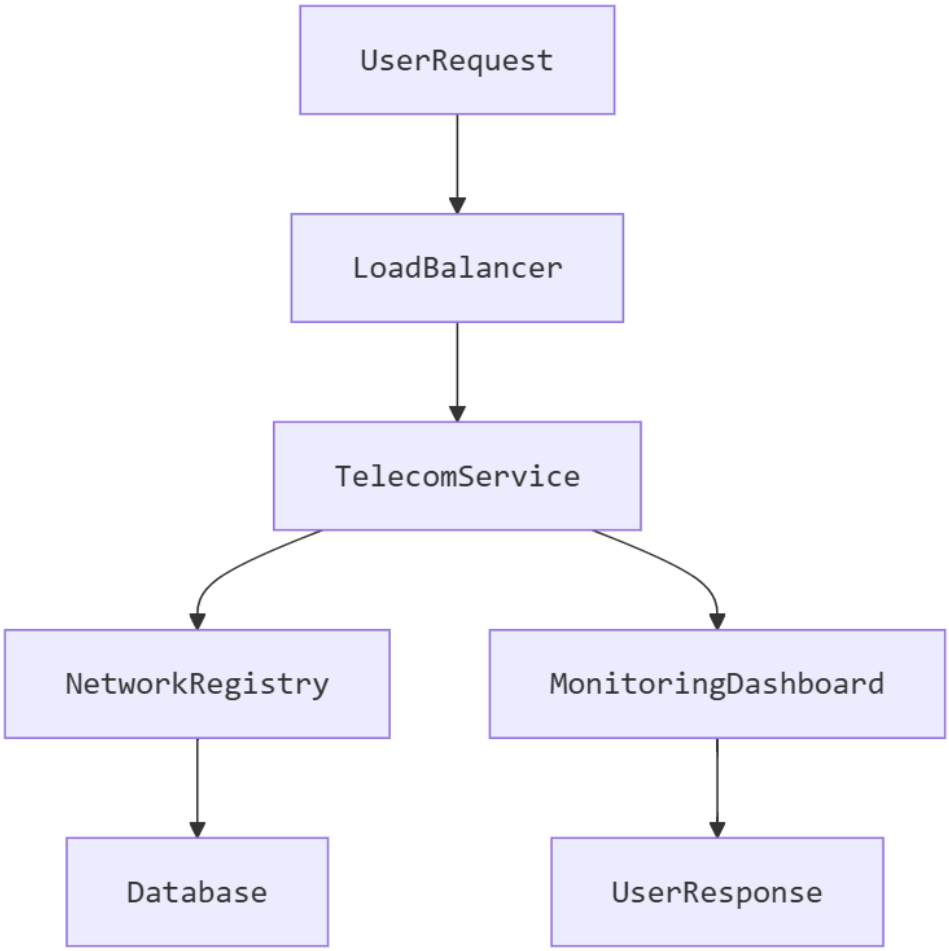
- Base Image: python:3.11-slim or alpine-based telecom binaries
- Exposed Ports: 5060 (SIP), 8080 (Dashboard), 3000 (Monitoring), etc.

2.2 Process Flow

End-to-End Provisioning Workflow



2.3 Information Flow



2.4 Components Design

Component 1 — Terraform (Infrastructure Provisioning)

Property	Value
Provider	AWS
Region	ap-south-1
Instance Type	t3.small
VPC	Custom or Default
Security Group	SIP + HTTP + SSH
AMI	Amazon Linux 2

Key Terraform Steps

- Initialize provider
- Create VPC + Subnets
- Create Security Groups
- Create EC2 Instances
- Output public/private IPs

Component 2 — Ansible (Configuration Management)

Configuration	Value
Inventory Source	Terraform output
Playbooks	install-docker.yml, deploy-services.yml
SSH Access	Private key from GitHub Secrets

Responsibilities

- Install Docker & dependencies
- Pull telecom service containers
- Start telecom network services
- Setup monitoring stack

Component 3 — Local Docker Telecom Network

Component	Port	Purpose
SIP Handler	5060	Call signalling
SMS Router	8081	Message routing
Registry Service	9000	Subscriber DB
Monitoring	3000	Grafana/Streamlit

Docker compose provisions:

- Telecom containers
- Shared overlay network
- Volume mounts
- Health checks

Component 4 — GitHub Actions (Automation Pipeline)

Step	Action
1	Checkout Repo
2	Install Terraform & Ansible
3	Run Terraform Init/Plan/Apply
4	Parse EC2 outputs
5	Run Ansible Playbooks
6	Validate services are running

Component 5 — AWS Infrastructure

Component	**Details**
EC2	t3.small, 2 instances
VPC	1 public subnet, IGW
Security Group	SIP/HTTP/SSH
IAM	Least privilege role

2.5 Key Design Considerations

API	**Method**	**Purpose**	**Auth**
/call/initiate	POST	Simulate call setup	Token
/sms/send	POST	Send SMS routing job	Token
/registry/query	GET	Subscriber info	None
/health	GET	Service health	None

Example Response:
200 OK
"status: healthy"

3 References

- Terraform Docs — developer.hashicorp.com
- Ansible Docs — docs.ansible.com
- Docker Docs — docs.docker.com
- AWS EKS/EC2 — docs.aws.amazon.com
- Telecom Standards — 3GPP TS Series

4 GitHub

https://github.com/atisha224/IaC-Provisioning-for-Telecomm-System